9.4 The Tangent Ratio

Geometry 9.4 · Student sheet

Name _____ Period ____ Date _____

I can…

- find the tangent of an acute angle in a right triangle.
- use tangent to find a missing leg of a right triangle.
- use an angle of elevation to solve real-life problems.

Vocabulary · fill in the missing word

Trigonometric ratio — a ratio of the lengths of two sides of a right _____

Tangent — tan A = (leg _____ ∠A) ÷ (leg adjacent to ∠A)

Angle of elevation — the angle from the _____ up to your line of sight

Part 1 · Discovery: same angle, same ratio

Draw a right triangle with a **35°** angle. Use the adjacent leg your teacher gives you. Measure the opposite leg.

Adjacent leg	Opposite leg	Opposite ÷ adjacent
Another pair:		
Another pair:		

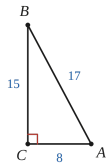
tan 35° on the calculator: _____ What do you notice?

Part 2 · Notes and examples (fill in each note, then work the example)

Tangent: tan A = _____ ÷ _____

Example 1

Find tan A and tan B. Write each as a fraction and as a decimal rounded to four places.



Same angle, same ratio: the tangent of an angle is the same in every _____ right triangle, no matter how big.

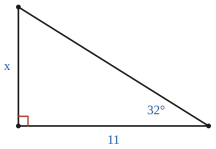
Example 2

Use a calculator to find tan 52° and tan 38°. Round to four places. What do you notice?

Finding a leg: write tan = opposite ÷ adjacent, put in what you know, and _____ for x. Keep the calculator in _____ mode.

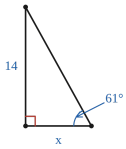
Example 3

Find the value of x. Round to the nearest tenth.



Example 4

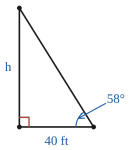
Find the value of x. Round to the nearest tenth.



Angle of elevation: the angle from the _____ up to your line of sight.

Example 5

You stand 40 feet from the base of a tree. The angle of elevation from the ground to the top of the tree is 58°. How tall is the tree? Round to the nearest tenth.



Part 3 · Partner practice

Partner 1

A right triangle has legs of 7 and 24 and a hypotenuse of 25. ∠A is across from the leg of 7. Find tan A and tan B as fractions and decimals.

Partner 2

Find the value of x. Round to the nearest tenth.

